Class: Algorithms & Data Structures Semester: Spring 2025

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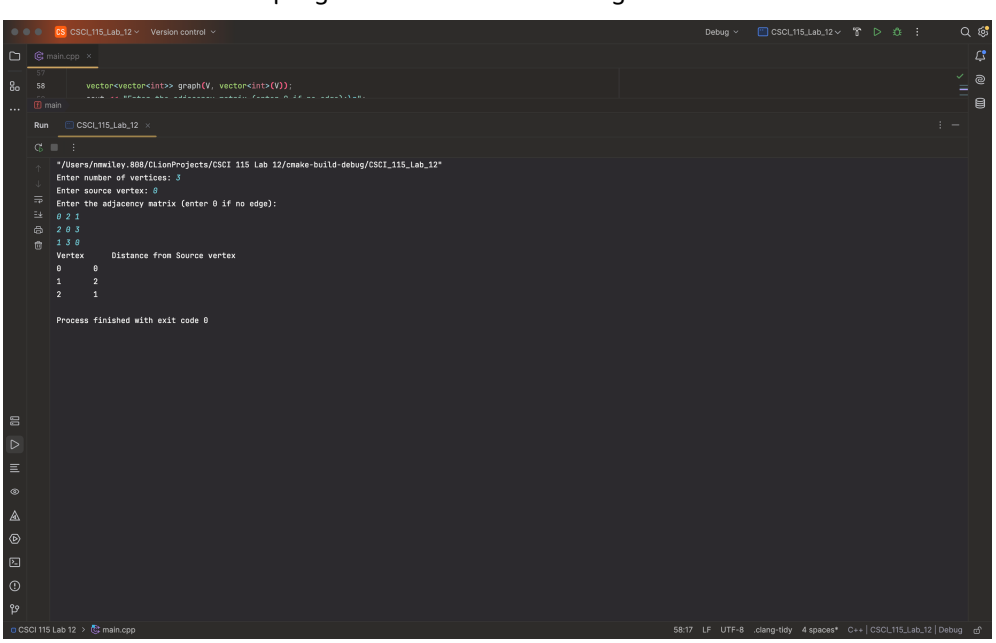
Lab Number - Section 09

CSCI 115 Lab 12; Dijkstra's Algorithm

- Statement of Objectives
  - The objective of this lab is to implement Dijkstra's algorithm in C++20 to find the shortest distances from a specified source vertex to all other vertices in a graph represented by an adjacency matrix. This lab provides experience in graph algorithms and helps understand how to calculate shortest paths efficiently. The report covers the design, implementation, testing, and analysis of the algorithm. This report details the experimental procedure, analysis, encountered challenges, and conclusions drawn from this experiment.

- Experimental Procedure
  - We implemented Dijkstra's algorithm in C++20, reading the number of vertices, the source vertex, and the adjacency matrix from user input. The algorithm uses an array to track visited vertices and an array to store the shortest distances. We tested the program using example graphs and verified the output by comparing it to expected results.

- Analysis
  - Below is a screenshot of the program results from running it:



- The output correctly shows the shortest distances from vertex 0 to all other vertices. For example, the shortest path from 0 to 1 is through edge 0-1 (weight 2), and to 2 is through edge 0-2 (weight 1).

- Encountered Problems
  - Initially, aligning the tabular output required fine-tuning with setw() to match the example format. Another challenge was handling disconnected nodes correctly, which we addressed by representing them with INF (infinity).

- Conclusions
  - From this lab, we learned how to:
    - Implement Dijkstra's algorithm using an adjacency matrix
    - Properly format tabular output in C++
    - debug and test graph algorithms with various input cases.

- References
  - Lab Lecture Slide on Dijkstra's algorithm